Class 9 Biology – Improvement in Food Resources (Annual Exam 2026)

Full Exam-Style Questions, Answers & Exact Keywords (NCERT + CBSE Aligned)

Crop Production and Management

Q1. Define sustainable agriculture. Why is it necessary for food security?

Answer: Sustainable agriculture is a farming practice that aims to meet current food requirements without harming the environment or reducing the ability of future generations to meet their needs. It is necessary for food security because it conserves soil fertility, water resources, and biodiversity while ensuring long-term crop productivity.

Keywords: sustainable agriculture, environmental protection, food security, future generations

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Q2. Differentiate between Kharif and Rabi crops. Give two examples of each.

Answer: Kharif crops are grown during the rainy season, sown in June–July and harvested in September–October, for example rice and maize. Rabi crops are grown during the winter season, sown in October–November and harvested in March–April, for example wheat and gram.

Keywords: Kharif, Rabi, rainy season, winter season, sowing, harvesting

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Q3. Distinguish between macronutrients and micronutrients required by plants. Give examples.

Answer: Macronutrients are nutrients required by plants in large quantities, such as nitrogen, phosphorus, and potassium. Micronutrients are required in very small quantities, such as iron, zinc, and manganese.

Keywords: macronutrients, micronutrients, large quantity, trace elements

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Q4. State four differences between manure and fertilizers. Why is excessive use of fertilizers harmful?

Answer: Manure is organic, improves soil structure, and releases nutrients slowly, whereas fertilizers are inorganic, do not improve soil structure, and release nutrients quickly. Excessive use of fertilizers is harmful because it reduces soil fertility, pollutes water bodies, and harms beneficial soil organisms.

Keywords: manure, fertilizers, soil fertility, pollution

Q5. What is organic farming? State its main characteristics.

Answer: Organic farming is a method of farming that avoids synthetic fertilizers and pesticides and relies on natural inputs like manure, compost, crop rotation, and biological pest control.

Keywords: organic farming, natural inputs, eco-friendly

Q6. Why is irrigation important? Describe two modern methods of irrigation.

Answer: Irrigation is important as it supplies water to crops at critical stages of growth. Drip irrigation supplies water directly to plant roots, reducing wastage, while river lift irrigation lifts water from rivers to irrigate fields in areas with irregular water supply.

Keywords: irrigation, water supply, drip irrigation, river lift

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Q7. Explain the following cropping patterns:

(a) Mixed cropping
(b) Inter-cropping
(c) Crop rotation

Answer: Mixed cropping involves growing two or more crops together to reduce risk of crop failure. Inter-cropping is growing different crops in alternate rows to improve resource use. Crop rotation is growing different crops in a fixed sequence to maintain soil fertility.

Keywords: mixed cropping, inter-cropping, crop rotation, soil fertility

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Q8. How do biotic and abiotic factors affect crop production?

Answer: Biotic factors such as pests and insects damage crops, while abiotic factors like heat, cold, salinity, and drought reduce crop yield by affecting plant growth.

Keywords: biotic factors, abiotic factors, pests, climate

Q9. What is hybridization? State its importance.

Answer: Hybridization is the crossing of two genetically different varieties to produce a new variety with desirable traits such as high yield and disease resistance.

Keywords: hybridization, high-yielding varieties, disease resistance

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Q10. Mention measures for safe storage of grains.

Answer: Safe storage of grains involves proper drying, fumigation, use of insecticides, moisture control, and storage in airtight containers.

Keywords: storage, fumigation, moisture control, pests

Animal Husbandry

Q11. What are the two main purposes of cattle farming? Differentiate between milch and draught animals.

Answer: The two main purposes are milk production and agricultural work. Milch animals are bred for milk production, while draught animals are used for labor such as ploughing.

Keywords: cattle farming, milch animals, draught animals

Q12. What is the lactation period? How can it be increased?

Answer: The lactation period is the time during which a dairy animal produces milk. It can be increased through proper nutrition, regular milking, and good health care.

Keywords: lactation period, milk production, nutrition

Q13. State desirable traits in poultry farming.

Answer: Desirable traits include high egg-laying capacity, disease resistance, fast growth, and low maintenance.

Keywords: poultry farming, disease resistance, productivity

Q14. Differentiate between broilers and layers.

Answer: Broilers are reared for meat and require protein-rich food, while layers are reared for eggs and require mineral-rich food.

Keywords: broilers, layers, meat, eggs

Q15. Differentiate between capture fishing, mariculture, and aquaculture.

Answer: Capture fishing involves catching fish from natural water bodies. Mariculture is fish farming in seawater, while aquaculture includes farming of aquatic organisms in freshwater and marine environments.

Keywords: capture fishing, mariculture, aquaculture

Q16. Explain composite fish culture.

Answer: Composite fish culture involves growing different species of fish together that feed at different levels of a pond. Its main advantage is efficient use of resources, while its disadvantage is difficulty in obtaining pure fish seeds.

Keywords: composite fish culture, resource utilization

Q17. Why is bee-keeping done? State qualities of Italian bees.

Answer: Bee-keeping is done for honey production and crop pollination. Italian bees produce more honey, are gentle, and have good breeding capacity.

Keywords: apiculture, Italian bees, honey

Q18. What is pasturage? How does it affect honey quality?

Answer: Pasturage refers to the availability of flowers for bees. It determines the quality, taste, and quantity of honey.

Keywords: pasturage, nectar, honey quality

Applied & Reasoning Questions

Q19. State desirable agronomic characteristics for cereal and fodder crops.

Answer: Cereal crops should have dwarf stature and high yield, while fodder crops should have tall growth and high biomass.

Keywords: agronomic traits, cereals, fodder

Q20. How did the Green Revolution and White Revolution improve food resources in India?

Answer: The Green Revolution increased food grain production using HYV seeds and fertilizers, while the White Revolution increased milk production through improved dairy farming.

Keywords: Green Revolution, White Revolution, food production

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✔️Prepared strictly as per **NCERT & CBSE Class 9 Biology (2026)** marking scheme